Video Style Transfer Evaluation - Synthetic Workflow Demonstration

Claim status: Exploratory
Date: 2026-11-08
Claim: Demonstrate a reproducible style-content-motion evaluation workflow without presenting synthetic clips as model benchmark evidence.

Executive verdict

SSIM ↑: Simulated balanced 0.9290 vs Simulated flicker 0.8111 (better; n=3).

LAB palette W1 ↓: Simulated balanced 9.0208 vs Simulated flicker 7.4201 (worse; n=3).

OFFSET 1 WARP RMSE ↓: Simulated balanced 31.70 vs Simulated flicker 47.77 (better; n=3).

OFFSET 1 FLOW EPE ↓: Simulated balanced 0.2438 vs Simulated flicker 0.5426 (better; n=3).

Runtime ↓: Simulated balanced 18.80 vs Simulated flicker 15.00 (worse; n=3).

Blinded VLM review: Simulated balanced has the highest mean overall visual quality (4.00/5 across 1 reviewer(s)); this is perceptual evidence, not benchmark ground truth.

Most frequent visual failure tag: detail loss (3 occurrence(s)).

The diagnostic style metric measures palette alignment only; texture and brushwork claims require perceptual evidence.

Evidence gates

Gate	Value	Status
Paired valid cases	3	Pass
Methods	3	Pass
Blinded VLM review	Present	Pass
Paper-grade perceptual metrics	Not detected	Not established
Claim status	Exploratory	Caveat

LAB palette distance and Farneback flow are deterministic diagnostic proxies. They are not RAFT, LPIPS, CLIP, FID, or ArtFID.

Quantitative results

Style

Method	LAB palette W1 (lower)	LAB chroma palette W1 (lower)
Simulated balanced	9.0208 ± 0.5445 (n=3)	3.1765 ± 1.3380 (n=3)
Simulated flicker	7.4201 ± 0.5347 (n=3)	1.8824 ± 1.0178 (n=3)
Simulated overlay	8.0416 ± 0.9411 (n=3)	4.6879 ± 0.4238 (n=3)

Content

Method	SSIM (higher)	PSNR (higher)
Simulated balanced	0.9290 ± 0.0165 (n=3)	22.58 ± 0.9018 (n=3)
Simulated flicker	0.8111 ± 0.0326 (n=3)	17.54 ± 0.3337 (n=3)
Simulated overlay	0.7153 ± 0.0149 (n=3)	14.90 ± 0.3931 (n=3)

Temporal and motion

Method	OFFSET 1 WARP RMSE (lower)	OFFSET 5 WARP RMSE (lower)	OFFSET 1 FLOW EPE (lower)	OFFSET 1 FLOW CORRELATION (higher)	Detrended luminance flicker (lower)
Simulated balanced	31.70 ± 6.7038 (n=3)	33.20 ± 5.2688 (n=3)	0.2438 ± 0.0558 (n=3)	0.4268 ± 0.2381 (n=3)	0.1035 ± 0.0619 (n=3)
Simulated flicker	47.77 ± 10.70 (n=3)	46.18 ± 10.55 (n=3)	0.5426 ± 0.3239 (n=3)	0.2267 ± 0.0923 (n=3)	12.43 ± 0.4689 (n=3)
Simulated overlay	63.87 ± 6.8744 (n=3)	63.70 ± 7.5960 (n=3)	0.1874 ± 0.0547 (n=3)	0.2662 ± 0.1339 (n=3)	0.0822 ± 0.0217 (n=3)

Cost

Method	Runtime (lower)	Peak GPU memory (lower)
Simulated balanced	18.80 ± 0.4000 (n=3)	12.70 ± 0.0000 (n=3)
Simulated flicker	15.00 ± 0.4000 (n=3)	9.8000 ± 0.0000 (n=3)
Simulated overlay	16.50 ± 0.4000 (n=3)	10.40 ± 0.0000 (n=3)

Blinded visual review

Method	Style adherence	Content preservation	Temporal stability	Artifact control	Overall visual quality
Simulated balanced	4.00/5	4.67/5	3.67/5	3.33/5	4.00/5
Simulated flicker	4.67/5	3.67/5	2.00/5	2.00/5	2.33/5
Simulated overlay	3.33/5	3.00/5	3.00/5	2.00/5	2.00/5

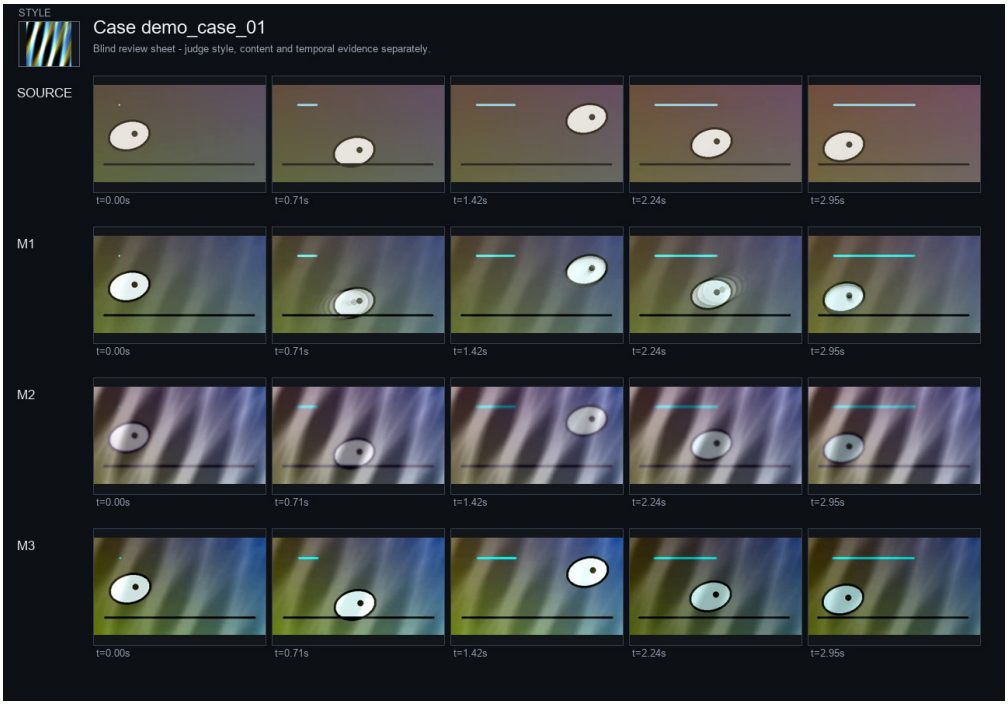
Failure tags: detail loss (3), texture sticking (3), flicker (3), edge ghosting (2), temporal jump (2), texture drift (1), geometry melting (1)

Reviewer records: 1. Scored case/method records: 9. VLM scores are perceptual evidence, not ground truth.

Per-case outcomes

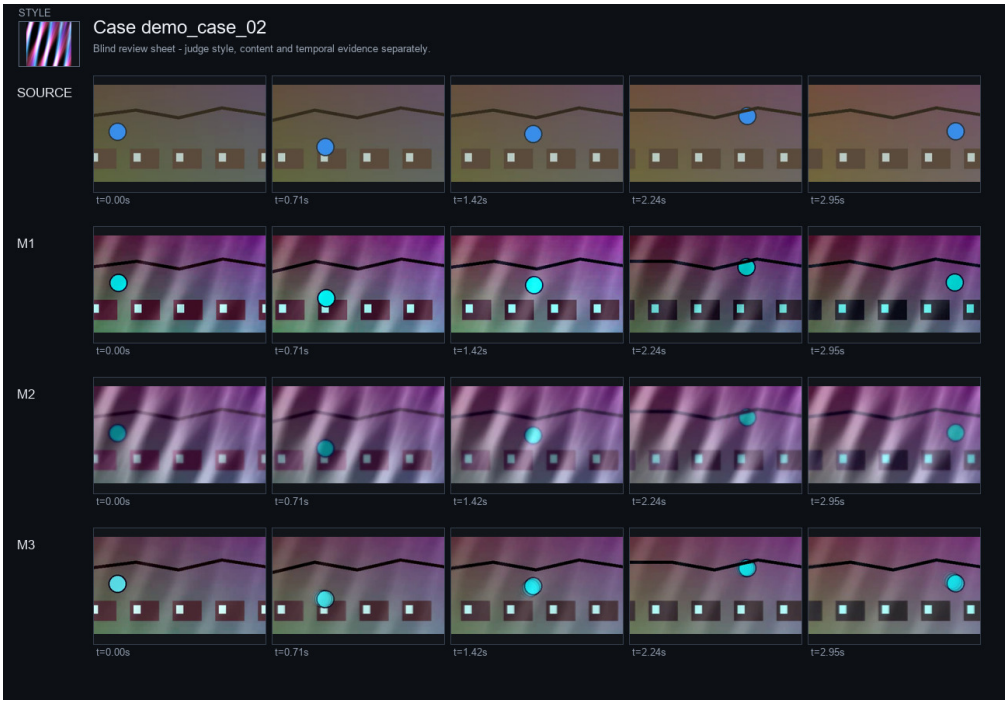
demo_case_01

Method	SSIM (higher)	Palette W1 (lower)	Warp RMSE d1 (lower)	Flow EPE d1 (lower)
Simulated balanced	0.9395	8.8136	28.61	0.2963
Simulated flicker	0.8270	7.3344	44.06	0.5494
Simulated overlay	0.7122	8.1708	65.43	0.1389



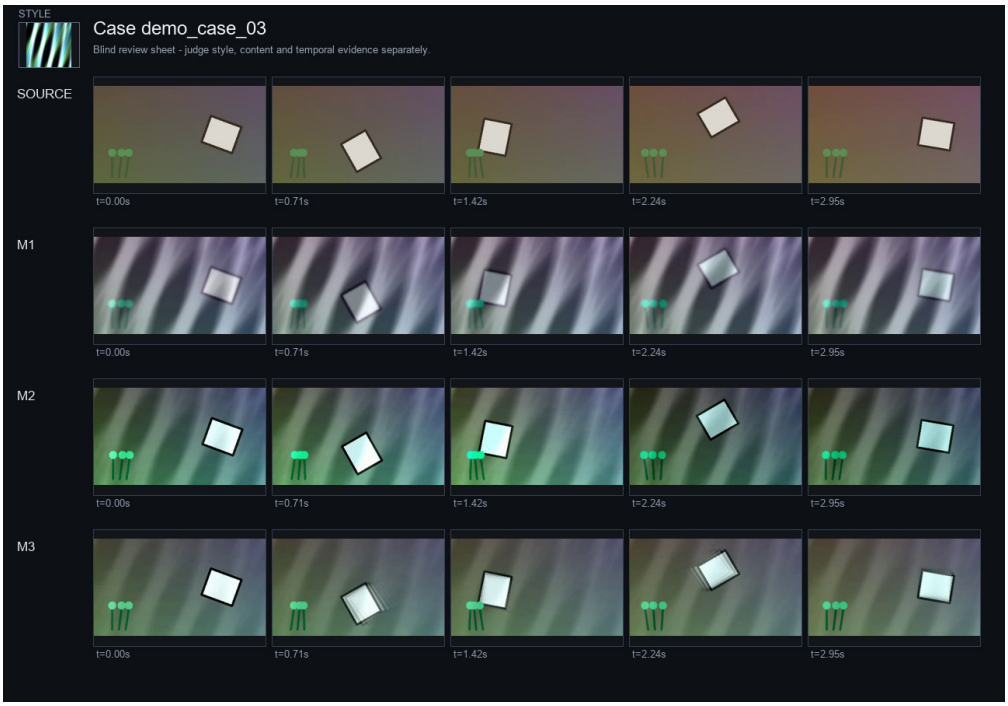
demo_case_02

Method	SSIM (higher)	Palette W1 (lower)	Warp RMSE d1 (lower)	Flow EPE d1 (lower)
Simulated balanced	0.9101	8.6103	39.39	0.2500
Simulated flicker	0.7736	6.9334	59.84	0.8630
Simulated overlay	0.7315	7.0425	69.83	0.2467



demo_case_03

Method	SSIM (higher)	Palette W1 (lower)	Warp RMSE d1 (lower)	Flow EPE d1 (lower)
Simulated balanced	0.9376	9.6385	27.09	0.1852
Simulated flicker	0.8327	7.9925	39.42	0.2154
Simulated overlay	0.7022	8.9114	56.35	0.1766



Threats to validity

- Content and style distributions may not represent the intended use.
- Low-level metrics are sensitive to colorization and compression.
- Farneback flow is not RAFT and can fail on large motion or occlusions.
- A single style image does not define a valid FID style distribution.
- VLM review can have position, content, and prompt bias.
- External baseline numbers copied from papers may not use the same settings.
- Fewer than five clips do not support a stable comparative confidence interval.

Recommendation

Use these results to validate the evaluation pipeline and choose the next experiment. Do not treat the synthetic run as model evidence.